

L1

CLASS X - SCIENCE

HOW DO ORGANISMS REPRODUCE ?

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Josh
Meter

What is Reproduction?

Reproduction is the biological process by which living organisms produce new individuals (offsprings) of their own species.

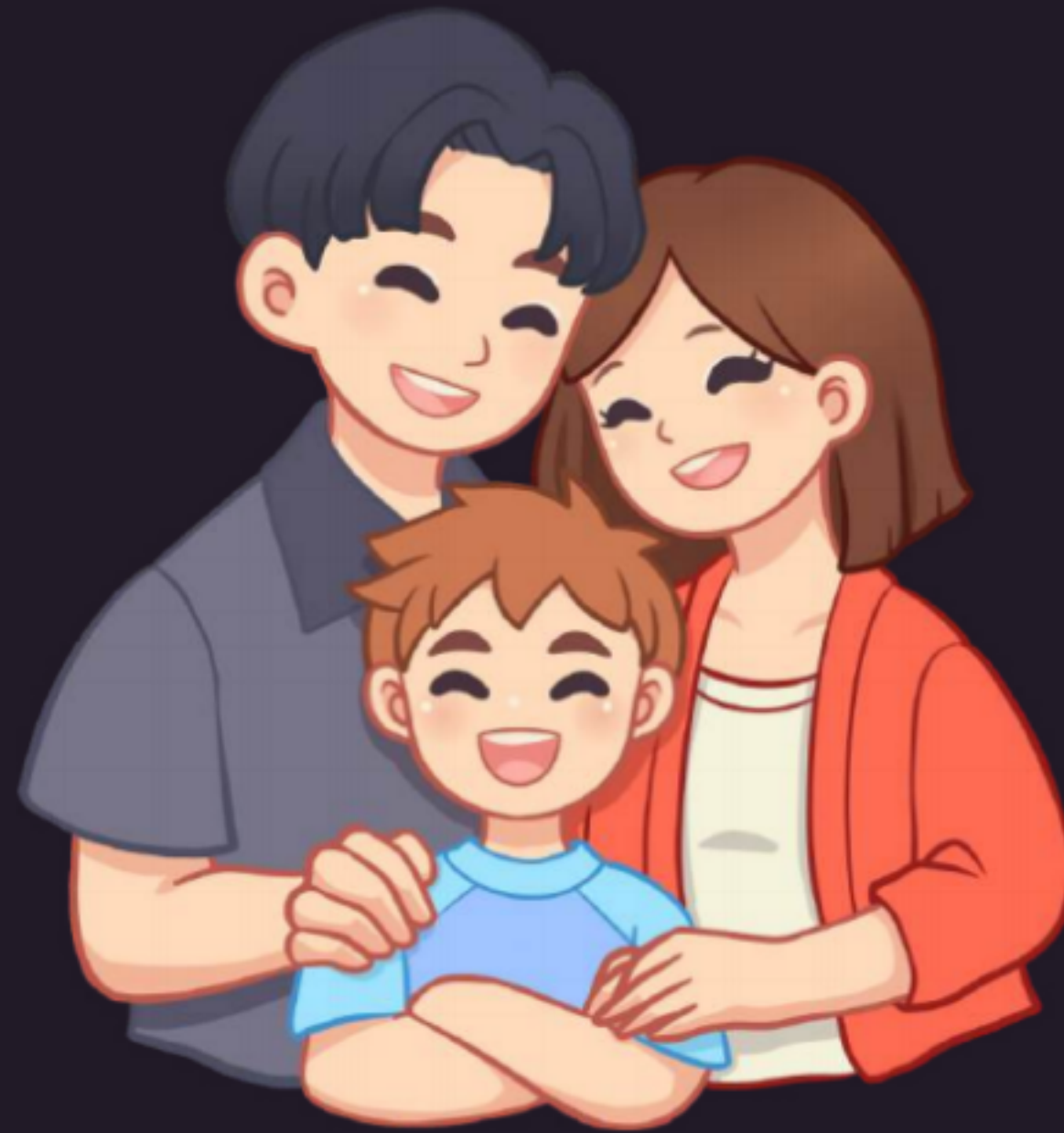
Q. Reproduction is one of the most important characteristics of living beings. Give three reasons in support of the statement.

Advantages of Reproduction

1. **Continuity of Species** – Ensures organisms of a species do not die out.
2. **Transfer of Characters** – Passes hereditary information (DNA) from parents to offspring.
3. **Population Continuity** – Maintains and increases population of organisms.
4. **Variations** – Provides small differences in DNA → useful for adaptation and evolution.
5. **Survival in Changing Environment** – Variations help species adjust to environmental changes.

Do organisms create exact copies of themselves?

Observe carefully :-

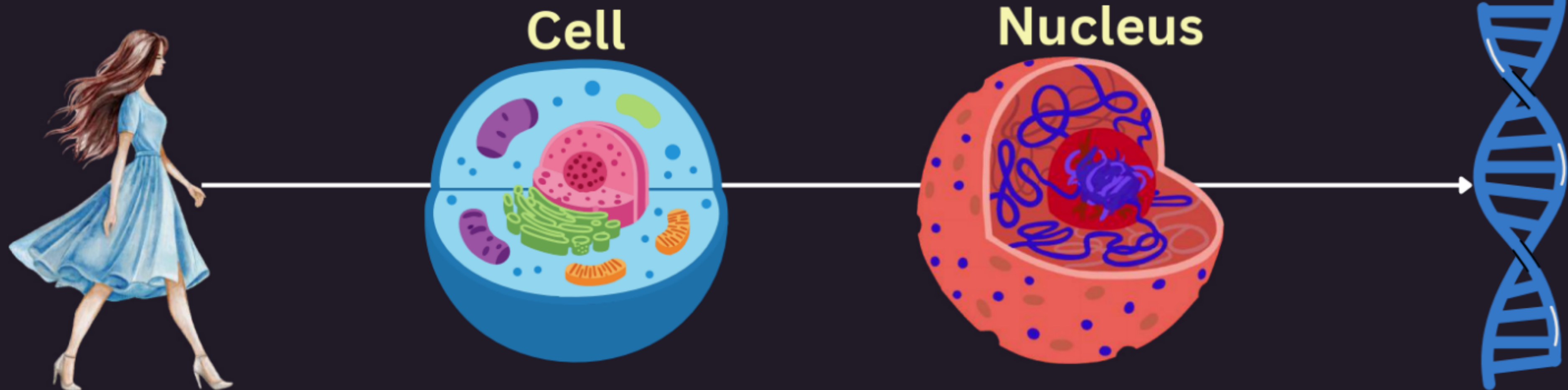


Offspring inherit many characteristics of their parents; they are similar but not identical.

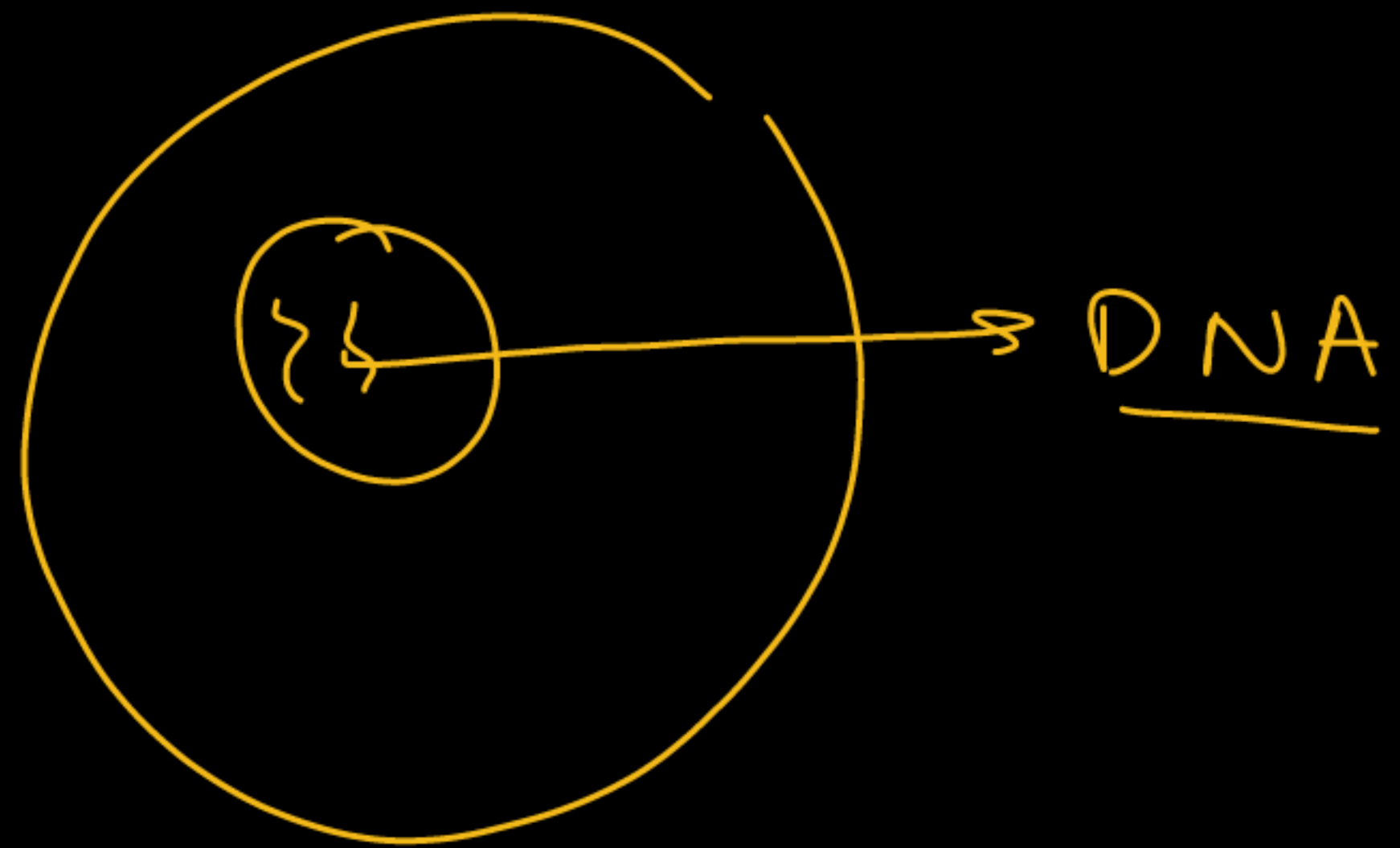


DNA - The blueprint of life

DNA (Deoxyribonucleic acid) is the genetic material present in the chromosomes inside the nucleus of a cell.



DNA stores genetic information and transfers it from one generation to the next.



Confused Between DNA, Chromosomes and Chromatin

DNA (Deoxyribonucleic Acid):



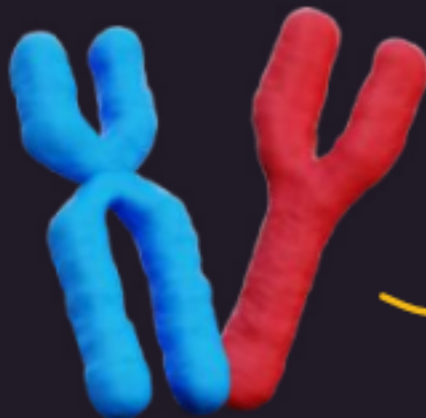
→ is a double-helical molecule present in the nucleus of a cell that carries hereditary information and controls the transfer of characters from parents to offspring.

Chromatin



→ is a thread-like structure made up of DNA and proteins found in the nucleus of non-dividing cells. It helps in packaging the DNA and carries the genetic material.

Chromosomes



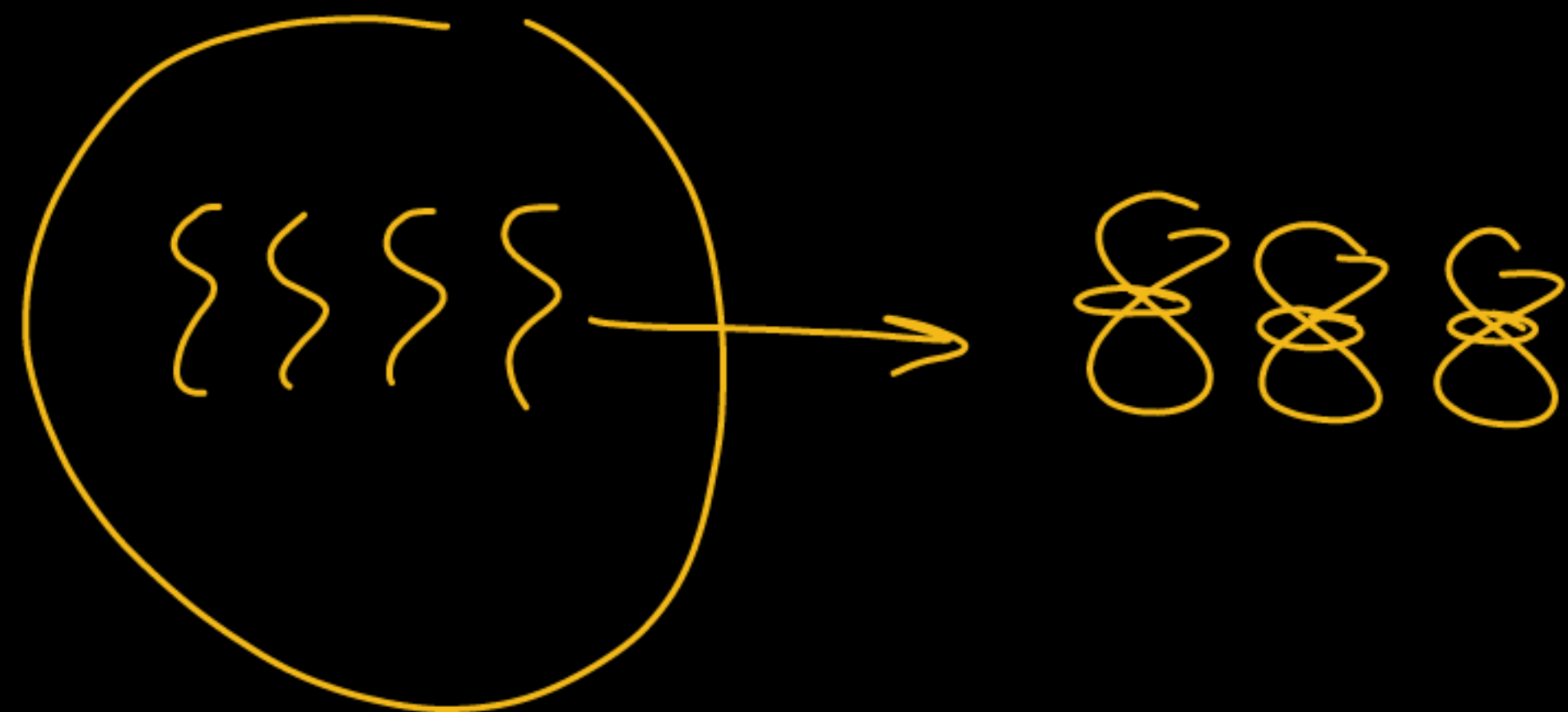
→ Chromosomes are condensed, rod-shaped structures formed from chromatin during cell division. They carry genes that transmit hereditary information from one generation to the next.



DNA

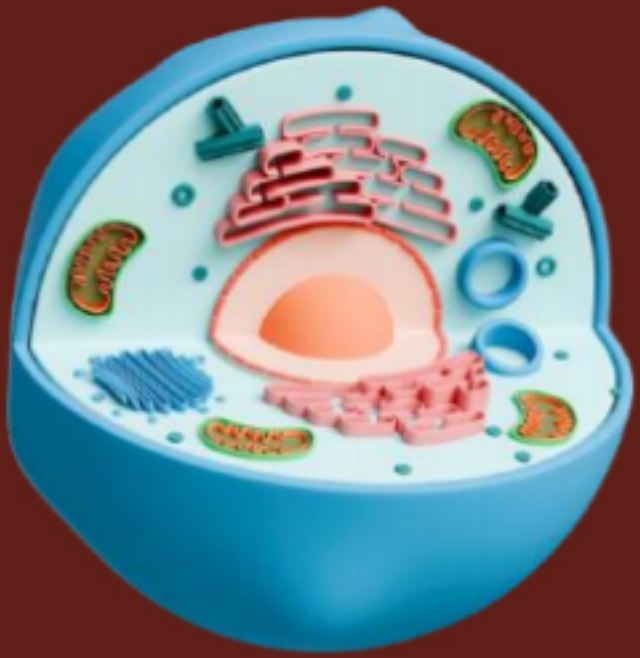
→ Chromatin

→ Chromosomes



DNA copying—the first step of reproduction

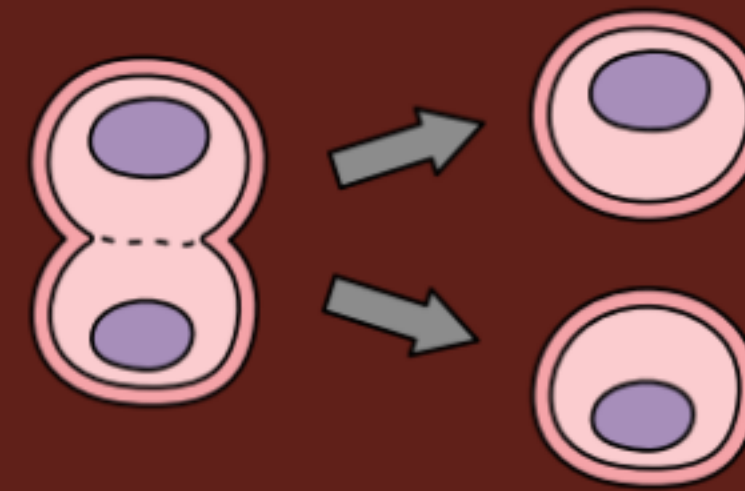
Before a cell reproduces, its DNA must be copied so that each new cell receives a complete set of genetic information.



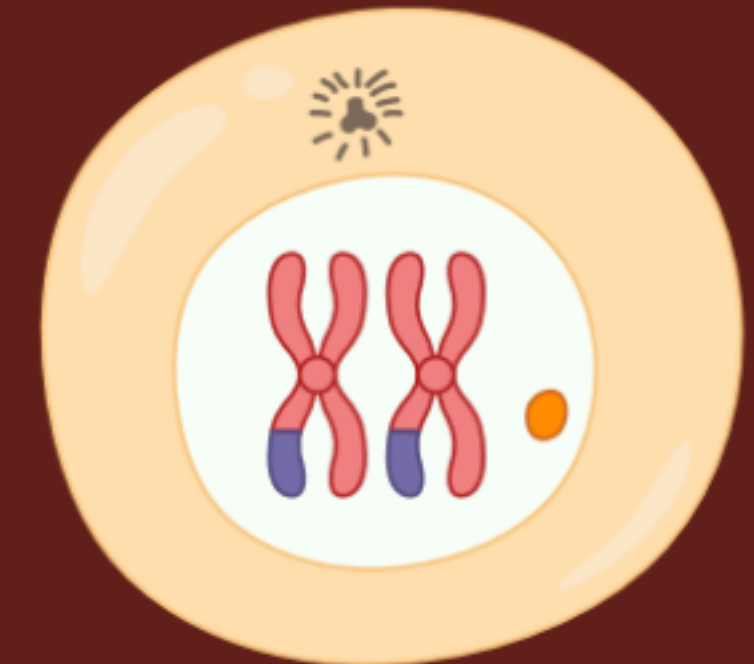
The original cell containing DNA molecule.



DNA copying: Two DNA copies are formed



Cell division: DNA copies separate into two daughter cells.



Daughter cells: each cell receives one DNA copy.

Variation



Sir, if DNA is copied before reproduction, then shouldn't every offspring be an exact copy of its parents?

Excellent questions! DNA copying is highly accurate, but it's not completely perfect. Tiny natural changes can occur during the copying process which causes Variations.

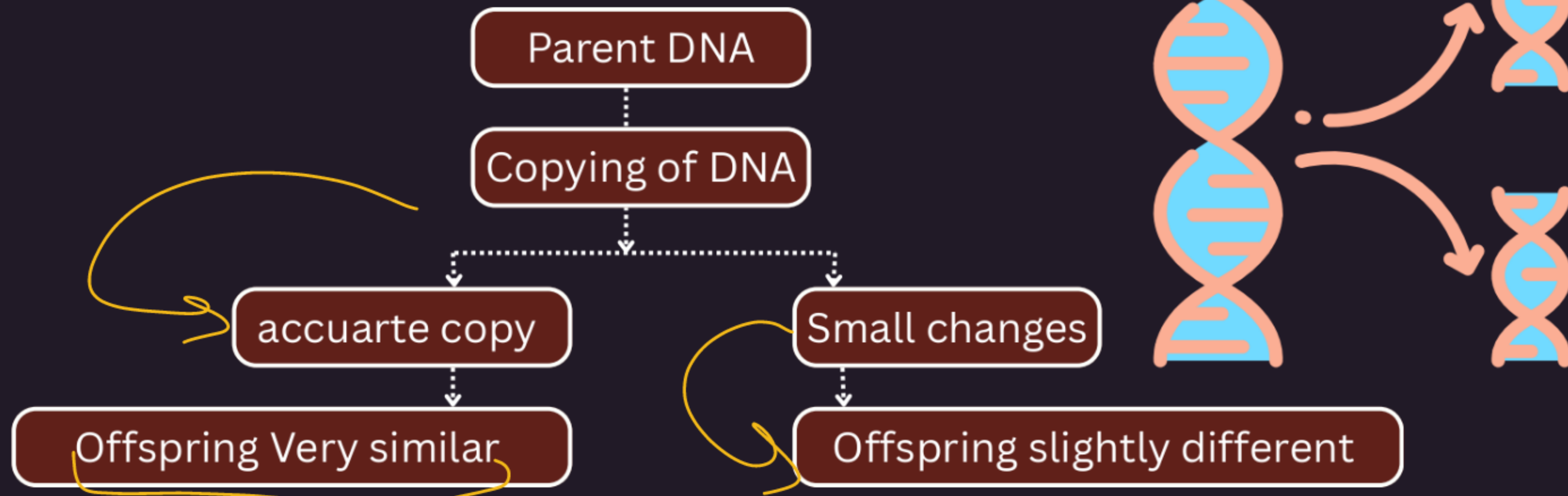
so, is that why no two individuals are exactly the same?

Exactly! These small differences are called variations.



How do variations arise?

DNA copying is highly accurate, but it is not completely error free. During copying, small changes may occur naturally, leading to variations in the offspring.



Importance of Variation

Ques: Explain how variation helps a population of bacteria to survive in changing environmental conditions such as an increase in temperature.



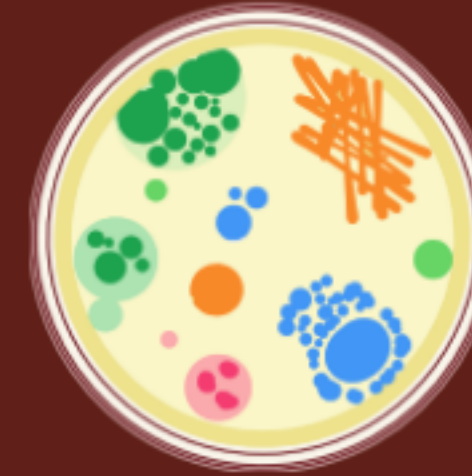
A popular of bacteria lives in moderate temperature water.

**Normal
environment**



Due to global warming, water temperature increases.

**Temperature
rises**



Most bacteria die.
A few grat resistant bacteria with useful variations survive.

**Only some
survive**



The surviving bacteria multiply and a new population develops. The species continues!

**species
continues**

Significance Of VARIATIONS

- **Survival and Adaptation:** Variations help organisms survive and adapt to changing environmental conditions.
- **Prevention of Extinction:** Variations increase the chances of survival of a species during sudden environmental changes.
- **Role in Evolution:** Useful variations may be inherited by the next generation and contribute to evolution over time.
- **Diversity Within a Species:** Variations in DNA create differences among individuals of the same species.

moth
(*Biston betularia*)



“Moth Ki Maut”

Not a useful Variation

Evolution of Human



Useful Variation

Check yourself

Qus:- Why can a species with greater variations have a better chance of survival when environmental conditions suddenly change ? Explain with an example.

Ans:- A species with greater variations has a better chance of survival when environmental conditions suddenly change because some individuals may possess variation that helps them adapt to the new conditions.

Example:- Some bacteria may have variations that make them resistant to antibiotics, allowing them to survive and reproduce.

Check yourself

Ques: A cell is preparing to reproduce. It first produces a copy of DNA and then produces additional cellular apparatus. Why is simply pushing one DNA copy out of the original cell not sufficient to produce a new organism/cell?

Ans: DNA alone can not form a new cell because it lacks the cellular machinery needed for life processes. Therefore, along with DNA copying the cell also makes extra cell parts and then divides into two cells.

Check yourself

Ques: What is the importance of DNA copying in reproduction? [NCERT]

Answer:

- DNA contains information for making proteins which determine the basic body design of an organism.
- DNA copying ensures that this genetic information is passed from parents to offspring.
- Thus, offspring maintain the basic features of their parents.
- Small variations may also arise during DNA copying.

Check yourself

Qus:- Which of the following is a source of variation during reproduction?

- a) DNA Copying
- b) Cell division
- c) Formation of new cell
- d) Protein formation

Check yourself

Ques. Read the statements carefully

- 1) Reproduction is not a essential process for an individual.**
- 2) Reproduction is essential for the survival and continuity of a species.** T
- a) Both statements are correct.
 - b) statement 1 is correct statement 2 is incorrect
 - c) statement 2 is correct statement 1 is incorrect
 - d) Both statements are incorrect.

Check yourself

Read the statements carefully:

- Statement I : Offspring inherit characteristics from their parents
- Statement II : Every offspring is an exact copy of its ~~parent~~.

Choose the correct option

- a) Both statements are true.
- ☒ b) Statement I is true but II is false
- c) statement I is false but II is true.
- d) Both are false

parent

Check yourself

Ques: Why is the variation benefits to the species but not necessarily for the individuals?

Ans. Variations help a species survive when environmental conditions change. Some individuals may possess variations that are favourable under the changed conditions and therefore survive and reproduce. However, a particular variation may not always be beneficial to an individual.

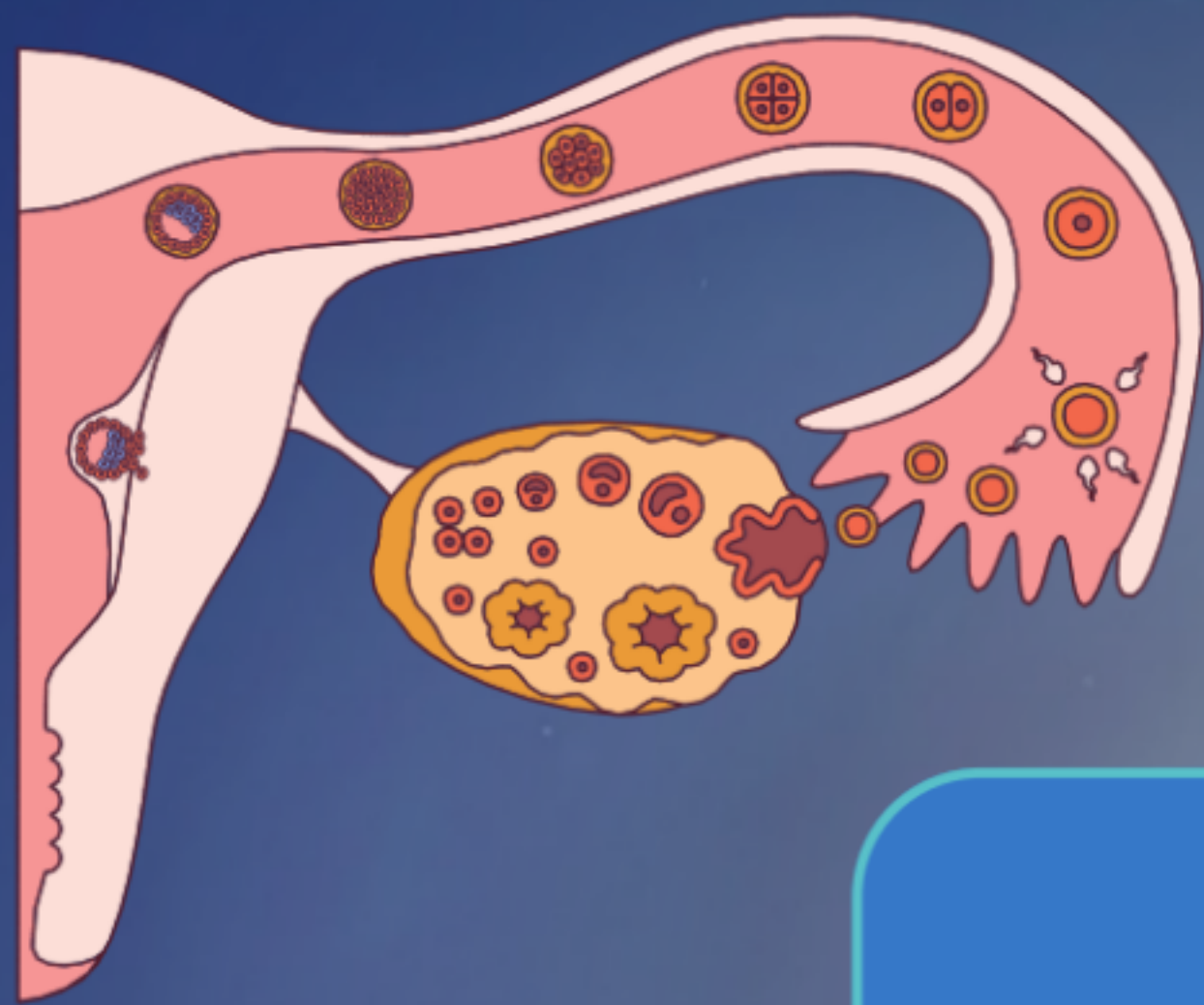
Check yourself

Ques: How does reproduction help in providing stability to population of species? [NCERT]

Answer:

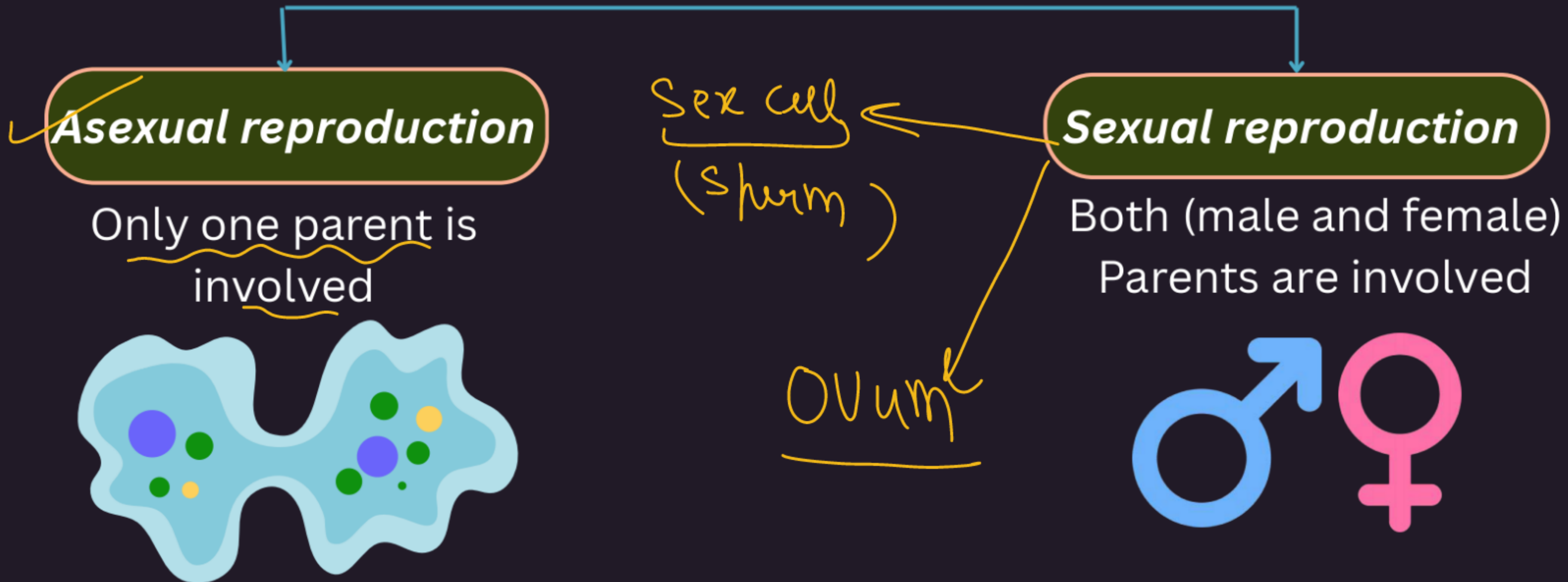
Reproduction maintains the continuity of a species by producing new individuals. The consistency of DNA copying maintains the basic body design of the species, while small variations may help the species survive changing environmental conditions.

Types of Reproduction



Types of Reproduction

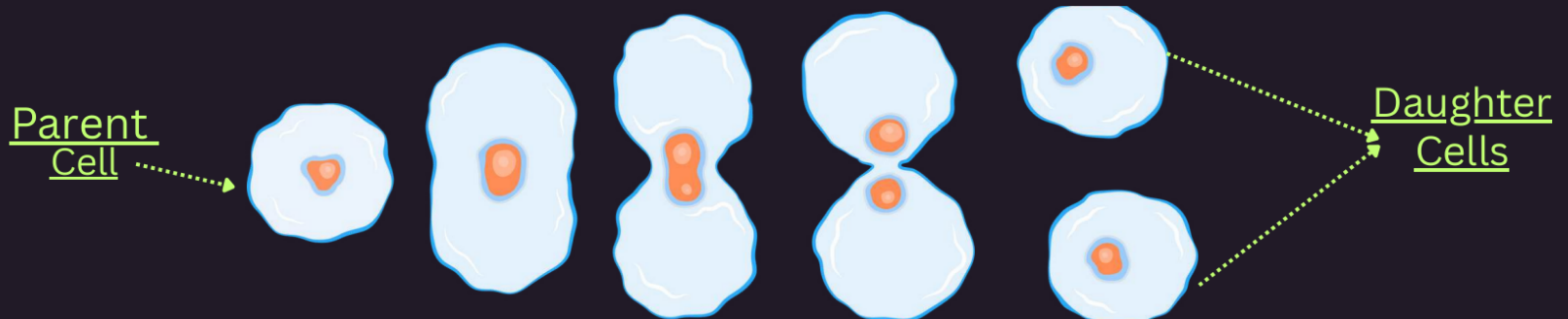
Living organisms reproduce through different methods depend on their body complexity.



Asexual reproduction

- Asexual reproduction is the mode of reproduction in which a single parent produces new individuals without formation or fusion of gametes.
- The offspring are generally very similar to the parent

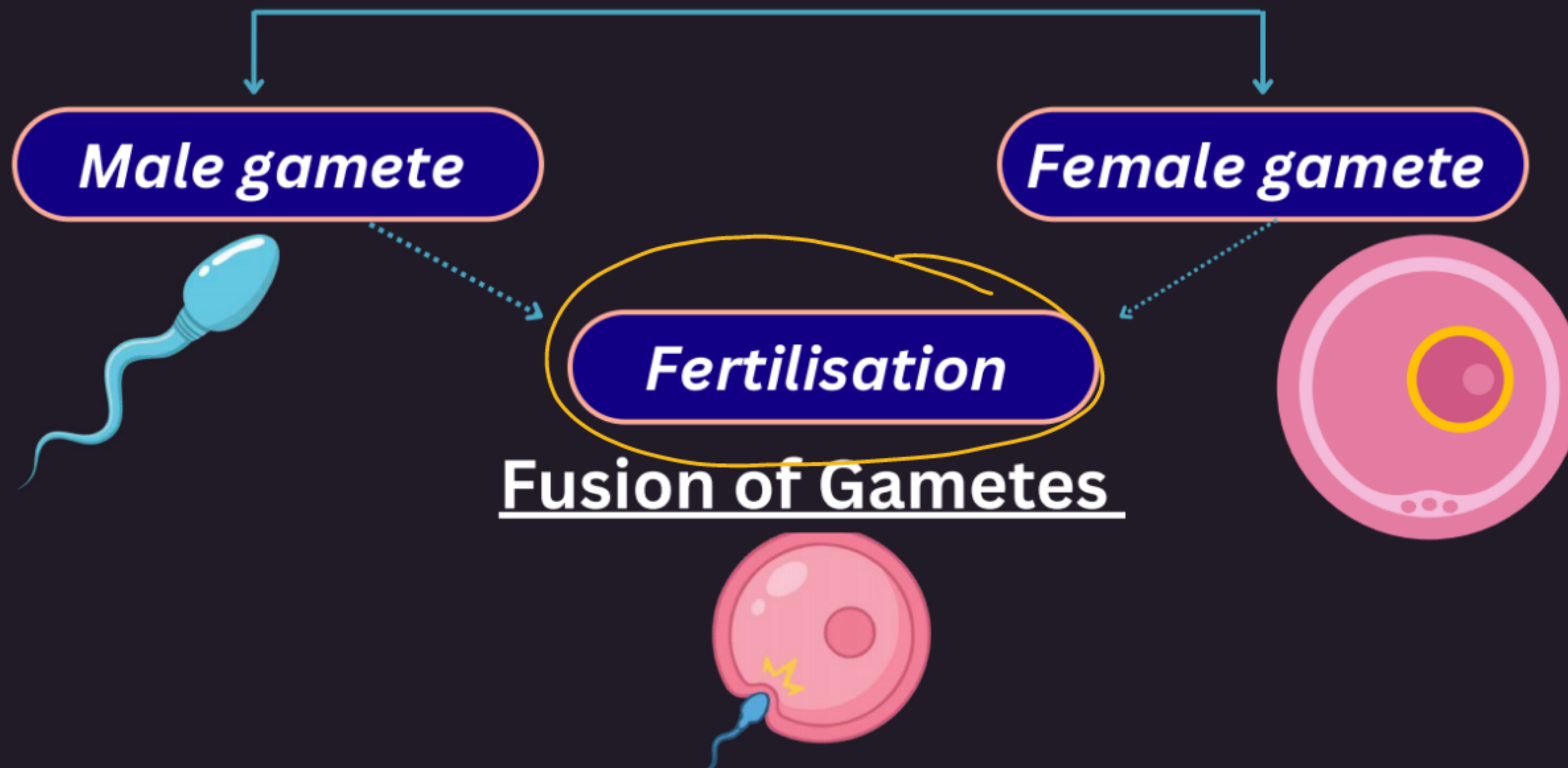
i.e., One parent cell dividing into two daughter cells.



Sexual Reproduction

Sexual reproduction is the mode of reproduction in which two parents are involved.

Fusion of gametes occur.



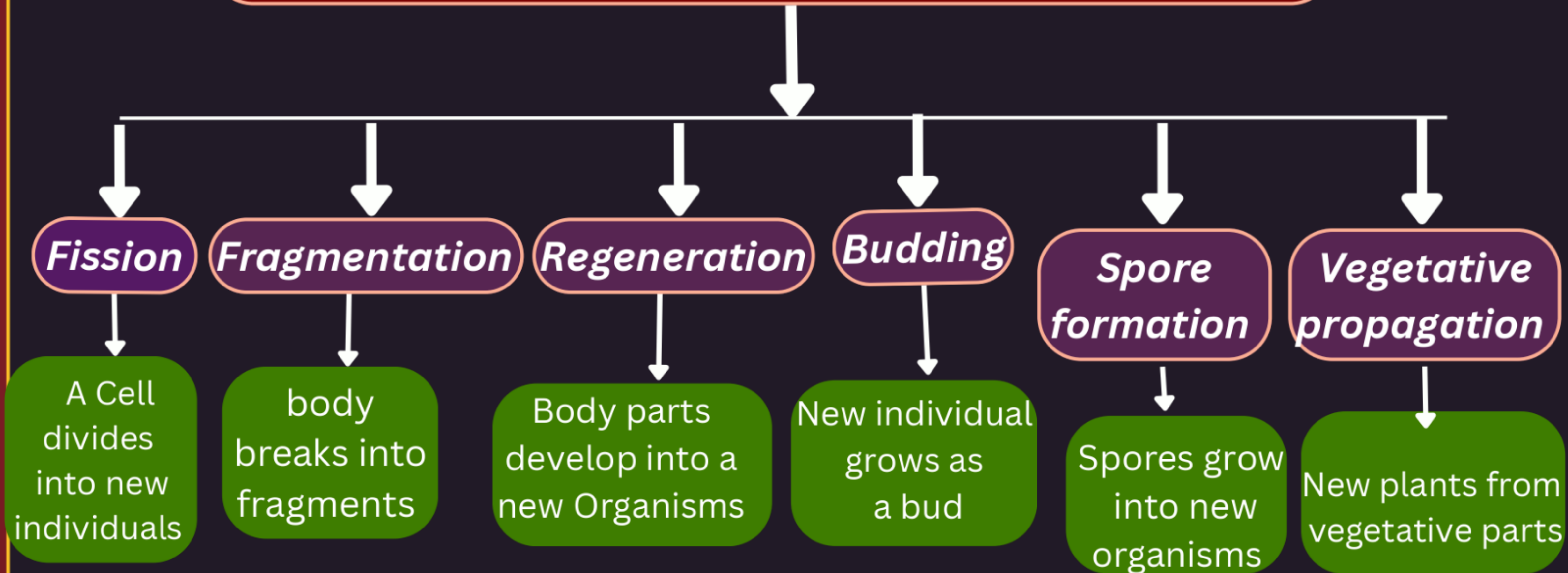
Asexual Vs Sexual reproduction



Q

<i>Asexual reproduction</i>	<i>Sexual Reproduction</i>
→ Involves one parent 	→ Involves two parents
<u>No fusion of gametes</u>	✓ Fusion of <u>male</u> and <u>female</u> gametes
<u>Offsprings are genetically very similar to parents</u>	✓ <u>Offspring shows greater variation</u>
<u>Faster process</u>	<u>Comparatively slower process</u>
Common in <u>unicellular</u> organisms or Some simple multicellular organisms	✓ Common in most higher plants and Animals
✓ Produces little variation	✓ Produces greater variation, improving adaptation

Types of asexual reproduction



2012, 2014, 2015, 2016, 2018, 2019, 2021

H W

Q. Assertion and Reasons

Assertion (A): Asexual reproduction produces genetically identical offspring.

Reason (R): DNA is copied from one parent with high accuracy.

- ☒ A. Both (A) and (R) are true and (R) is the correct explanation of (A).
- B. Both (A) and (R) are true and (R) is not the correct explanation of (A).
- C. (A) is true, but (R) is false.
- D. (A) is false, but (R) is true.

2013, 2015, 2017, 2019, 2020, 2022, 2023

Q. Which mode of reproduction is better – asexual reproduction or sexual reproduction? Why?

- Sexual reproduction involves fusion of male and female gametes, leading to variation in offspring.
- These variations help species to adapt to changing environments and promote evolution.
- In contrast, asexual reproduction produces genetically identical offspring, so they are less adaptable to environmental changes.



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